

A Knowledge Graph Approach to the Preservation of Creative Code-Based Works with Large Language Models: A Pilot Study Using Max-Based Compositions for the Laptop Orchestra of Louisiana¹

Lindsey Hartman
College of Art and Design
Louisiana State University
Baton Rouge, LA, United States
Lhartm1@gmail.com

Abstract:

Creative code-based works depend on interconnected software, hardware, operating environments, external libraries, and tacit performance knowledge that are rarely documented in a single location. To bring these distributed materials and dependencies into a centralized preservation infrastructure, this pilot study develops and evaluates an AI-assisted system for three Max-based compositions created by the Laptop Orchestra of Louisiana between 2014 and 2016. The system combines a structured file repository with a Neo4j knowledge graph designed to represent the works' descriptive, technical, provenance-related, and performance-related information and relationships. A locally executed Python workflow extracts file inventories and dependency evidence from preserved work folders and Max patch files, while large language model tools support schema development, metadata preparation, unstructured document ingestion, and natural-language querying.

The study evaluates both the usefulness and limitations of these tools within a cultural heritage preservation workflow. Structured graph data supported accurate retrieval of missing dependencies, version constraints, hardware requirements, and other digital preservation risks. However, automated entity extraction produced duplicate entity nodes and disconnected document chunks, while the natural-language agent struggled to retrieve contextual information from those documents. These findings demonstrate the importance of traceable evidence, human validation, and preservation-specific data modeling when applying AI systems to complex cultural computing collections. The resulting prototype provides a foundation for expanding the repository and further evaluating human-AI approaches to the preservation of code-based creative works.

Workshop Presentation:

For this workshop, I will focus on how the pilot repository and knowledge graph bring technical and contextual knowledge together for three legacy Max-based compositions from the Laptop Orchestra of Louisiana. Useful applications and limitations of AI-supported tools across the workflow will be identified, with attention to where practitioner judgment remained necessary.

1. **Related publication:** L. Hartman, "Preserving Code-Based Creative Works as Cultural Computing Records: A Human-AI Knowledge Graph Approach," Doctoral Consortium paper, in *2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, Dallas, TX, USA, 2026, forthcoming.